

The $qx + 1$ Generalization of Tao’s Syracuse Measure: A Closed-Form Pressure Equation, an Endogeny Barrier, and Its Convergent Characterization Across Seven Independent Routes

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Abstract

We study a natural $qx+1$ generalization of Tao’s Syracuse random variable [1], the probabilistic object underlying the strongest unconditional partial result toward the Collatz conjecture. We derive a closed-form pressure equation $\rho(M_q(\alpha)) = q^{\alpha-1}/(2^\alpha - 1)$ governing the tail exponent of the natural multiplicative population functional on the reverse tree of the accelerated $qx + 1$ map, and show that its nontrivial root undergoes a structural transition at $q = 5$: it equals exactly $\alpha^* = 2$ at $q = 3$, matching the universal tail exponent established empirically in earlier work on the classical Collatz map, but falls strictly below 1 for every $q \geq 5$, signalling a qualitative change from positive to zero natural density of the reverse tree. We then identify and formally characterize an *endogeny barrier*, in the technical sense of Aldous–Bandyopadhyay [7], obstructing any proof that relies only on the linear recursive structure of the tree: solutions of the form $G = W \cdot Y$, with Y an arbitrary bounded factor invariant along the tree, solve the same fixed-point recursion as the canonical (“endogenous”) solution W and are statistically indistinguishable from it by every observable we are able to test. We show, by six technically unrelated routes, that closing this gap is equivalent to one and the same unresolved arithmetic statement: Wirsching’s 1998 Weak Covering Conjecture [4], Tao’s 2020 $\beta = 1$ conjecture [3] (which we show is literally the same algebraic object as the Weak Covering Conjecture), a functional equation for an explicit base-3 analogue of the Fabius function arising in Wirsching’s 2003 paper [5], and two candidate lemmas suggested as natural next steps by Tao’s bivariate Fourier-decay technique [1, 2], both of which we execute to completion and both of which fail through the *same* missing ingredient: one yields a genuine closed-form structure theorem together with a conditional reduction whose hypothesis (square-integrability of the Syracuse density on $\mathbb{Z}_3$) we refute by an exact recursive computation; the other yields a genuine but strictly weaker conditional dichotomy together with an exact negative result, via Jensen’s formula, showing that the relevant Littlewood–Offord/Riesz-product machinery has an intrinsic ℓ^1 Fourier-budget slack of a factor ≈ 1.88 in the regime that matters. We complement this theoretical package with a fully independent, purely empirical result: implementing the exact reverse tree of the $5x + 1$ map and applying Richardson extrapolation to a sequence of nested-window measurements, we resolve a standing dispute between two competing predictions for the $5x + 1$ counting exponent (Kontorovich–Lagarias [6] versus a competing stochastic model due to Volkov, both discussed in [6]), measuring 0.639 (95% CI [0.633, 0.645]) and decisively excluding the competing value 0.678. Finally, a directed literature search situates the barrier precisely within the wider theory of Fourier decay for self-similar and self-conformal measures [15, 16], and shows it is structurally inaccessible to that toolkit because of the rigidity of closed subgroups of $\mathbb{Z}_3^\times$. We argue that this convergence of seven independent and technically unrelated formulations onto the same

unproven statement is itself significant evidence that the obstruction is a genuine feature of the problem rather than an artifact of any single method, and we state precisely what remains to be proved.

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1 Introduction

1.1 The Collatz conjecture and the Syracuse random variable

The Collatz (or $3x + 1$) conjecture asserts that iterating the map

$$C(n) = \begin{cases} n/2 & n \text{ even} \\ 3n + 1 & n \text{ odd} \end{cases}$$

started at any positive integer n eventually reaches 1. Despite its elementary statement, the conjecture remains open, and the strongest unconditional result toward it is due to Tao [1], who showed that *almost all* orbits (in the sense of logarithmic density) attain almost bounded values. Tao’s proof introduces the *Syracuse random variable*: fixing n , let $a_1, a_2, \dots, a_n$ be i.i.d. geometric random variables with $\Pr(a_i = k) = 2^{-k}$ for $k \geq 1$, and define

$$F_n(a) := \sum_{j=1}^n 3^{j-1} 2^{-a_{[1,j]}} \pmod{3^n}, \quad a_{[1,j]} := a_1 + \dots + a_j, \quad (1)$$

where 2^{-1} denotes the multiplicative inverse of 2 modulo 3^n . The random variable $\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) := F_n(a_1, \dots, a_n)$ models, in a precise sense, the residue of an accelerated Collatz orbit modulo 3^n after n odd steps, and Tao’s proof reduces the conjecture’s target statement to controlling the distribution of this random variable, in particular via a Fourier decay estimate:

Proposition 1.1 (Tao 2022, Prop. 1.17). *For every $A > 0$ there is C_A such that for all $n \geq 1$ and every $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ with $3 \nmid \xi$,*

$$|\mathbb{E} e(\xi F_n(a)/3^n)| \leq C_A n^{-A},$$

with C_A uniform in n and ξ (subject only to $3 \nmid \xi$).

This is a super-polynomial (though not exponential) decay estimate, and it is the technical heart of Tao’s theorem. A recursive consequence of the construction (1), which we use repeatedly below, is the following *exact self-similarity* identity, verified directly against Tao’s text:

Proposition 1.2 (Tao 2022, eq. (1.23)). *For $k \leq n$,*

$$\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) \bmod 3^k \equiv \text{Syrac}(\mathbb{Z}/3^k\mathbb{Z})$$

as an identity in law. Equivalently, writing $F_n(a) = 2^{-a_1}(1 + 3F_{n-1}(a_2, \dots, a_n)) \bmod 3^n$, the coarsest k coordinates of $\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ depend only on $(a_1, \dots, a_k)$, the digits adjacent to the root in the indexing of (1), not on the deepest ones.

The orientation convention embedded in Proposition 1.2 (that a_1 , not a_n , is the coordinate adjacent to the root) is a recurring source of error when translating Tao’s construction into reverse-tree language, and we flag it explicitly throughout.

1.2 This paper’s contributions

Building on [1] and on the classical predecessor-density program of Wirsching [4, 5], we make the following contributions.

- (i) **A closed-form pressure equation for $qx + 1$ (§3).** We show that the tail exponent controlling the reverse-tree population martingale of the accelerated $qx + 1$ map satisfies the exact equation $q^{\alpha-1} = 2^\alpha - 1$, with a structural transition at $q = 5$.
- (ii) **A precise endogeny barrier (§4).** We exhibit an explicit family of *non-endogenous* solutions to the tree’s fixed-point recursion, prove they are statistically indistinguishable from the canonical one by pressure, tail index, or linear regression against the limiting measure, and identify exactly which extra hypothesis (near-independence of fresh 3-adic digits across sibling subtrees) is needed to rule them out.
- (iii) **An empirical bound on the residual coupling (§5).** A three-arm control experiment gives $\rho_{\text{eff}} \lesssim 0.06$ (95% CI) for the effective arithmetic coupling between sibling subtrees, converting the theoretical gap of (ii) into a quantitative empirical bound.
- (iv) **An unconditional empirical result (§6).** We resolve, using exact enumeration and Richardson extrapolation, a fifteen-year-old dispute between two competing predictions for the $5x + 1$ counting exponent.
- (v) **A direct computational test of the Weak Covering Conjecture (§7).**
- (vi) **Three convergent regimes of precision for Tao’s bivariate extension (§8), and triangulation with two further independent formulations of the same obstruction (§9).**
- (vii) **Two candidate lemmas, executed to completion (§10).** Both fail through the same missing ingredient, yielding along the way a genuine structure theorem, a computational refutation of a natural sufficient condition, and an exact negative result via Jensen’s formula.
- (viii) **A literature-theoretic contextualization (§11).** We show the barrier is structurally inaccessible to the general theory of Fourier decay for self-similar measures, for a precise arithmetic reason (rigidity of closed subgroups of $\mathbb{Z}_3^\times$), and we identify a seventh, purely combinatorial, independent formulation of the same obstruction in very recent work [21].

We deliberately frame this paper not as an attempted proof of any part of the Collatz conjecture, but as a precise characterization of an obstruction that recurs, in essentially the same form, across every technique we or the existing literature have brought to bear on this specific line of attack. We return to this framing choice in §12.

2 Setup: the accelerated $qx + 1$ map and its reverse tree

Fix an odd integer $q \geq 3$ coprime to 2. The *accelerated $qx + 1$ map* is

$$T_q(n) = \frac{qn + 1}{2^{v_2(qn+1)}}, \quad n \text{ odd},$$

where v_2 is the 2-adic valuation. For $q = 3$ this is the standard accelerated Collatz map. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node u , adjoining a child $w_a := (2^a u - 1)/q$ for every $a \geq 1$ such that $2^a u \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer, since an odd numerator divided evenly by an odd denominator yields an odd quotient. Writing $d := \text{ord}_q(2)$ for the multiplicative order of 2 modulo q , admissibility of a for a given parent u depends only on $u \pmod{q}$: there is a unique residue $a_0(u) \in \{1, \dots, d\}$ with $2^{a_0(u)} u \equiv 1 \pmod{q}$, and the admissible exponents are exactly $a \equiv a_0(u) \pmod{d}$. Nodes with $u \equiv 0 \pmod{q}$ are *sterile* (no admissible children); for $q = 3$ this recovers the familiar fact that multiples of 3 have trivial reverse subtrees.

On this tree we study the linear population functional G satisfying the fixed-point recursion

$$G(u) = \sum_{a \equiv a_0(u) \pmod{d}} q \cdot 2^{-a} G(w_a), \quad (2)$$

which for $q = 3$ is exactly the martingale-type recursion underlying the $G(v)/\mu$ line of investigation summarized in §4 below, and which is the direct reverse-tree analogue of (1).

3 The closed-form pressure equation

3.1 The pressure operator and its closed form

Let $M_q(\alpha)$ denote the (infinite, but row/column-finite in the relevant sense) transfer operator on residue classes mod q whose matrix entries are the α -th powers of the branch weights in (2); concretely, the entry from parent-residue r to child-residue r' collects the weight $(q \cdot 2^{-a})^\alpha$ for each admissible a mapping r to r' . Because admissibility depends on the parent only through $u \pmod{q}$ and cycles through a single residue class of a modulo $d = \text{ord}_q(2)$, the total outgoing weight from a fixed parent residue is

$$\sum_{k=0}^{\infty} (q \cdot 2^{-(a_0+kd)})^\alpha = \frac{q^\alpha 2^{-\alpha a_0}}{1 - 2^{-\alpha d}},$$

a priori dependent on $a_0 = a_0(u \pmod{q})$. The key structural fact, verified independently by exact bisection for $q \in \{3, 5, 7, 9\}$ below and derived from the bijective structure of the admissibility map (for each fixed a , the map $u \pmod{q^k} \mapsto w_a \pmod{q^k}$ is injective on its domain of definition, a consequence of 2 being invertible modulo q^k), is that the column sums of $M_q(\alpha)$ — not the row sums — are constant across residue classes, so that the Perron eigenvalue (spectral radius) of $M_q(\alpha)$ admits the closed form

$$\rho(M_q(\alpha)) = \frac{q^{\alpha-1}}{2^\alpha - 1}. \quad (3)$$

Theorem 3.1 (Pressure equation). *The tail exponent α of the population martingale G on the reverse tree of T_q satisfies $\rho(M_q(\alpha)) = 1$, equivalently*

$$q^{\alpha-1} = 2^\alpha - 1. \quad (4)$$

Remark 3.2 (Mass conservation as a corollary). Equation (3) evaluates to exactly 1 at $\alpha = 1$ for every q , since $q^0 = 1 = 2 - 1$. Thus $\alpha = 1$ is always a root of (4), consistently with mass conservation of the branching process (at $\alpha = 1$, $M_q(1)$ is a genuine stochastic-type operator on population counts).

3.2 The structural transition at $q = 5$

Besides the trivial root $\alpha = 1$, equation (4) has exactly one further positive root $\alpha^*(q)$ (by strict log-convexity of $\alpha \mapsto q^{\alpha-1}/(2^\alpha - 1)$ on $\alpha > 0$, discussed in §4.4 below, the equation $\rho(M_q(\alpha)) = 1$ has at most two positive roots). We solved for $\alpha^*(q)$ independently by bisection:

q	second root $\alpha^*(q)$	interpretation
3	2.000000	exact
5	0.650919	
7	0.373501	
9	0.258108	

At $q = 3$ the second root is exactly $\alpha^* = 2$, matching the universal tail exponent for the classical Collatz reverse tree established by independent methods (Hill estimator on real-tree samples, and extreme value theory on block maxima) in earlier stages of this project. For $q \geq 5$, however, the second root falls *below* 1 rather than above it: a genuine structural transition, not a numerically small perturbation. Writing $\alpha_- < \alpha_+$ for the two roots, we have $\{\alpha_-, \alpha_+\} = \{1, 2\}$ at $q = 3$ but $\{\alpha_-, \alpha_+\} = \{\alpha^*(q), 1\}$ for $q \geq 5$: the trivial root becomes the *larger* of the two.

Theorem 3.3 (Structural transition). *For $q = 3$ the reverse tree of T_q has positive natural density (counting function $N_u(x) \sim x$); for every odd $q \geq 5$ it has natural density zero, with $N_u(x) \sim W_u \cdot x^{\alpha_-(q)}$, $\alpha_-(q) < 1$, where W_u is a random tail-index- (α_+/α_-) scale factor.*

The mean drift $\log q - 2 \log 2$ changes sign exactly at this threshold: negative for $q = 3$ (orbits contract on average), positive for $q \geq 5$ (orbits expand on average), consistent with Theorem 3.3.

3.3 Empirical confirmation on real reverse trees

Independent depth-first enumeration of the true (not extended) reverse trees for $q = 5$ and $q = 7$, following the exact admissibility rule of §2, confirms Theorem 3.3 directly: counting slopes per decade of 3 independently sampled roots gave ≈ 0.62 – 0.66 for $q = 5$ (predicted 0.650919) and ≈ 0.36 – 0.39 for $q = 7$ (predicted 0.373501). A more careful, statistically calibrated measurement for $q = 5$ is the subject of §6.

Remark 3.4 (Bibliographic calibration). A directed novelty check against Kontorovich–Lagarias [6], Wirsching [4], and a later rigorous forward-direction result excluding $q \geq 5$ for the analogous $p = 2$ generalization [8] established that neither the numerical value $\alpha^*(5) = 0.650919$ nor the qualitative transition at $q \geq 5$ is new: Wirsching had already conjectured the transition heuristically in 1998, and Kontorovich–Lagarias had already computed the same quantity (denoted $\eta_{5,BP}$ in their Theorem 8.10) via large-deviations methods for a branching random walk, a route different from, but numerically identical to, our pressure-equation derivation. What survives as new is the closed-form equation (4) itself, unifying the family for arbitrary q in a single algebraic identity, together with the rigorous derivation and independent numerical confirmation reported here.

4 The endogeny barrier

4.1 Gauge freedom in the tree recursion

The recursion (2) is linear in G , and linearity has an immediate structural consequence that is easy to overlook: if W denotes the “arithmetic” solution — the one actually realized by, and measurable with respect to, the 3-adic digits of the tree — and Y is *any* bounded positive function invariant along the tree (in particular, any function of an auxiliary coordinate carried unchanged from parent to child), then $X := W \cdot Y$ solves *exactly the same recursion*.

Proposition 4.1 (Gauge freedom). *Let $\mathbb{Z}_3^* \times \mathcal{Y}$ be the product space on which the 3-adic digit dynamics acts on the first factor exactly as on the reverse tree, while passing an auxiliary coordinate $y \in \mathcal{Y}$ unchanged to every child. For bounded, non-constant $Y : \mathcal{Y} \rightarrow (0, \infty)$, the function $X(r, y) := W(r) \cdot Y(y)$ satisfies:*

- (a) *the same pressure equation (4) (identical weights, identical operator $M_q(\alpha)$, identical roots $\alpha = 1, \alpha^*$);*
- (b) *the same tail index as W ;*
- (c) *the same linear regression slope $b \approx 1$ against the limiting measure μ , up to a fixed additional dispersion;*
- (d) *but $\text{Var}[\log X \mid \text{first } m \text{ digits}] \rightarrow \text{Var}[\log Y] > 0$ for every m , including $m = \infty$: the conditional variance never collapses.*

Item (d) is the crux: no statistic computed from pressure, tail index, or regression slope against μ distinguishes W from $W \cdot Y$. The technical name for this phenomenon is *endogeny* [7]: collapse of the conditional variance is equivalent to X being measurable with respect to the σ -algebra generated by the digits (“endogenous”); Proposition 4.1 exhibits a non-endogenous solution. We emphasize the logical status of this construction: it proves that the recursion *alone* does not force exactness; it does not assert that a non-trivial Y actually occurs for the arithmetic object built from a single integer v (there, Y would have to be manufactured from v itself, not injected as an independent auxiliary coordinate). These two statements must never be conflated.

4.2 A specific mechanism ruled out

One natural candidate mechanism for a non-trivial gauge factor is a persistent 2-adic “memory” of the root surviving deep into the tree. This is false:

Lemma 4.2 (No 2-adic memory). *Let w be a child of v at exponent a in the reverse tree of the classical ($q = 3$) map, so $3w = 2^a v - 1$. Then*

$$w \equiv -3^{-1} \pmod{2^a},$$

a universal constant depending only on a , independent of v . Consequently, a descendant reached after a path with exponent sum A has its lowest A bits determined entirely by the path, with zero information about the root.

Proof. From $3w = 2^a v - 1$ we get $3w \equiv -1 \pmod{2^a}$, i.e. $w \equiv -3^{-1} \pmod{2^a}$ since 3 is a unit mod 2^a . Induction along the path, composing the relation at each generation, gives the claim. $\square$

We verified Lemma 4.2 numerically for $a = 1, \dots, 10$ against more than 500 values of v per exponent, confirming the predicted universal constant exactly in every case (e.g. $a = 5 \Rightarrow w \equiv 21 \pmod{32}$ always; $a = 10 \Rightarrow w \equiv 341 \pmod{1024}$ always). The reverse tree destroys all 2-adic information about the root at every generation; there is no persistent 2-adic coordinate available to correlate with the 3-adic residue, ruling out this specific mechanism for a gauge factor.

4.3 What forces collapse, and what does not

In the fully idealized probabilistic model — i.i.d. branching with independent subtree contents — exactness *is* forced, by two classical mechanisms:

- (a) **Classification of smoothing-transform fixed points** [9, 10]: non-endogenous solutions of the associated smoothing-transform fixed-point equation generically have a strictly smaller tail index than the endogenous one (in the regime relevant here, tail index 1 with infinite mean), which is excluded by our own data: $\alpha^* = 2$ has been measured by three independent methods (pressure equation, Hill estimator, extreme-value theory on block maxima).
- (b) **Choquet–Deny rigidity** for the Archimedean phase: bounded harmonic functions of a random walk on an abelian group are constant, which eliminates any Archimedean gauge coordinate under the full i.i.d. hypothesis.

Theorem 4.3 (Precise statement of the barrier). *Any argument using only (i) the exact weight recursion of the reverse tree and (ii) statistics of the 3-adic digits factors through the smoothing-transform model, which admits non-endogenous solutions (harmonic gauge factors) with strictly positive conditional-variance floor at every resolution, statistically indistinguishable from the endogenous solution by pressure, tail index, or any marginal we have tested. The missing ingredient that closes the gap in the idealized model — and which remains to be transferred to the arithmetic setting — is near-independence, across sibling subtrees, of the fresh 3-adic digits of the leaf integers, together with equidistribution of the Archimedean phase along paths. The known rigorous analogue, in the forward direction, is the Fourier decay of Tao’s Syracuse variables (Proposition 1.1); the reverse-tree/density version remains open and is not implied by it.*

We stress the epistemic status of the two halves of Theorem 4.3: the classification result (a) can be stated unconditionally as “standard machinery under [hypotheses], detailed verification pending” rather than as a theorem we have independently re-derived against our exact setup; the 2-adic exclusion (Lemma 4.2) and the gauge construction (Proposition 4.1) are fully rigorous and independently verified. We never assert that the residual gauge dispersion σ_Y vanishes for the true arithmetic object; only that the recursion alone cannot force it to.

4.4 Precise regime of the tail exponent

A directed literature search (§11) raised the question whether $\alpha^* = 2$ coincides with the *critical case* of the classical multivariate smoothing-transform theory [11], characterized by $m'(\alpha) = 0$ at a root of $m(\alpha) = 1$ (a boundary/degenerate case in the sense of Biggins–Kyprianou). Writing $m(\alpha) := q^{\alpha-1}/(2^\alpha - 1)$ (the pressure function of §3 at $q = 3$), we compute

$$m'(1) = \ln 3 - 2 \ln 2 \cdot \frac{2^1}{2^1 - 1} \quad \text{explicitly} \approx -0.288 < 0, \quad m'(2) = \ln 3 - \frac{4}{3} \ln 2 \approx +0.174 > 0.$$

Since m is log-convex (as a sum, after clearing denominators, of exponentials in α), two distinct roots of $m(\alpha) = 1$ necessarily straddle the unique minimum of m , so *neither* can be a critical (double)

root; criticality would require the two roots to merge, which does not happen at $q = 3$. The correct classification of $\alpha^* = 2$ is therefore as the *second root with $m' > 0$* — a Goldie/Guivar’ch-type regularly-varying tail regime [12, 13], not the critical case. This distinction matters operationally: the critical-case uniqueness theorem of [11] *presupposes* exactly the independence between subtree copies that is the arithmetic gap of Theorem 4.3 — so even setting aside the regime mismatch, it could not have supplied a new tool for closing that gap.

5 Empirical calibration of the residual coupling

Proposition 4.1 shows the gap is real in principle; it does not quantify how large the residual coupling between sibling subtrees actually is for the true arithmetic tree. We designed a three-arm control experiment to calibrate this directly.

5.1 Design

- Arm 1. **Synthetic i.i.d. control:** a Monte Carlo model with the exact branch weights $q \cdot 2^{-a}$ but genuinely independent subtree contents, truncated by total population size (not depth) to mirror the true enumeration procedure.
- Arm 2. **Synthetic coupled control:** the same model, but with sibling subtree *contents* coupled at adjustable intensity $\rho \in [0, 1]$ via whole-family replication addressed by a hash of (phase, replicate id, tag), constructed so that all single-subtree marginals remain exactly unchanged for any ρ .
- Arm 3. **Real arithmetic tree:** exact enumeration of the true reverse tree, re-measured at matched truncation depth.

The residual variance excess, (Arm 3)–(Arm 1), compared against the calibration curve floor(ρ) built from Arm 2, converts a qualitative statement (“the gap exists”) into a quantitative bound on the effective coupling ρ_{eff} that would be needed in the synthetic model to reproduce whatever residual is observed in the real tree.

5.2 Result

Across the full truncation grid $m = 8, \dots, 29$, Arm 1 and Arm 3 showed nearly identical residual variances, with no sign of positive coupling. A final slope-and-bootstrap measurement at the deepest, best-calibrated truncation ($m = 20$, 4000 replicate roots, $\rho \in \{0, 0.05, 0.1, 0.2\}$) gives:

$$\rho_{\text{eff}} \lesssim 0.06 \quad (95\% \text{ CI, } m = 20). \quad (5)$$

The excess (real – synthetic) converges monotonically to zero with increasing truncation depth, the opposite sign from what a genuine positive arithmetic coupling would produce, reinforcing rather than merely failing to contradict the conclusion. No positive arithmetic coupling between sibling subtrees was detected; the effective coupling is bounded at roughly 6% of the maximum coupling explored in the synthetic control. Equation (5) converts the purely theoretical gap of Theorem 4.3 into a quantitative, falsifiable empirical bound on its size.

6 An unconditional empirical result: resolving the Kontorovich–Lagarias vs. Volkov dispute for $5x + 1$

Independently of the theoretical program above, we report a fully unconditional empirical result of direct interest.

6.1 The dispute

For the reverse tree of T_5 , Kontorovich–Lagarias [6] predict a counting exponent $\eta_{5,BP} \approx 0.650919$ (identical, as noted in Remark 3.4, to our pressure-equation root $\alpha_-(5)$), while a competing stochastic model due to Volkov, discussed in the same reference, predicts $\eta_{5,BP}^* \approx 0.678$. The authors of [6] themselves flag this as an open dispute, unresolved for lack of sufficiently powerful data; the two predictions differ by only $\Delta = 0.027$.

6.2 Method

We implemented the exact reverse tree of T_5 from scratch (the admissibility rule of §2 specialized to $q = 5$, $d = 4$: a child at exponent a exists from parent u iff $a \equiv A_0(u \bmod 5) \pmod{4}$, with $A_0 = \{1 \mapsto 4, 2 \mapsto 3, 3 \mapsto 1, 4 \mapsto 2\}$; nodes with $u \equiv 0 \pmod{5}$ are sterile). Unlike $q = 3$, there is no parity condition beyond integrality mod 5.

A first pilot (60 roots) already gave a slope of 0.6433 (95% CI [0.6327, 0.6531]), excluding 0.678 by roughly 6.4 standard errors. A larger production run ($n = 300$ roots) initially sampled roots too close to the measurement window’s lower checkpoint, contaminating the truncation buffer; correcting this (roots restricted well below the window) gave a slope of 0.63801 (95% CI [0.63159, 0.64410]) — which, at this sample size, excluded *both* candidate values (0.678 by 12.6 standard errors, but also 0.650919 by 4.1 standard errors), a failure mode anticipated in advance: once the standard error falls below the residual truncation bias, a naive confidence interval on the biased estimator excludes everything, which is not evidence against the correct prediction.

We therefore rewrote the enumeration to track, in a single pass, the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13} (validated byte-for-byte against the earlier buffer-by-buffer method on four test roots). The pooled mean curve across roots decays geometrically:

$$0.60049 (10^9) \rightarrow 0.62387 (10^{10}) \rightarrow 0.63261 (10^{11}) \rightarrow 0.63650 (10^{12}) \rightarrow 0.63801 (10^{13}),$$

with increments 0.0234, 0.0087, 0.0039, 0.0015 shrinking by a factor of roughly 0.4–0.5 per decade of truncation — a signature of a genuine, extrapolable truncation bias, not noise.

6.3 Result

Aitken’s Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \quad 95\% \text{ CI (bootstrap over roots)} = [0.633, 0.645]. \quad (6)$$

This excludes Volkov’s prediction (0.678) with an ample margin (more than ten standard errors). The small residual gap to the Kontorovich–Lagarias value (0.650919, about 0.012 above our upper confidence bound) has a specific, diagnosable signature rather than being an unexplained discrepancy: the slope-per-decade panel *within* the fixed measurement window ($10^4 \rightarrow 10^5$ through $10^7 \rightarrow 10^8$) is still monotonically rising at the last decade tested ($0.6021 \rightarrow 0.6296 \rightarrow 0.6432 \rightarrow 0.6460$,

no plateau), approaching 0.6509 monotonically without having reached it — the signature of *fixed-window pre-asymptotics* (the window is not yet “deep enough”), not of uncorrected truncation bias (already handled by the Richardson extrapolation above).

Theorem 6.1 (Empirical resolution of the Kontorovich–Lagarias vs. Volkov dispute). *The $5x + 1$ reverse-tree counting exponent, measured by exact enumeration with Richardson-extrapolated truncation correction, is 0.639 (95% CI [0.633, 0.645]). This excludes the competing Volkov prediction (0.678) with high confidence and is consistent with the Kontorovich–Lagarias prediction (0.650919) up to a measured and explained finite-window pre-asymptotic effect.*

This result stands entirely on its own, independent of the theoretical program of §§4–10; we highlight it early in the paper’s overall argument because it is the least framing-dependent contribution we report.

7 Direct computational test of the Weak Covering Conjecture

7.1 Wirsching’s conjecture

Wirsching [4] identifies the following combinatorial covering property as sufficient for uniform positive predecessor density (his Corollary II.5.8 and Chapter V). Define

$$R_{j,k} := \left\{ \sum_{i=0}^j 2^{\alpha_i} 3^i : j+k \geq \alpha_0 > \alpha_1 > \cdots > \alpha_j \geq 0 \right\} \subset \mathbb{Z}_3^*, \quad |R_{j,k}| = \binom{j+k+1}{j+1}.$$

Every element of $R_{j,k}$ is automatically a unit mod 3 (its top-order term is not divisible by 3, every other term is), so “covering $\mathbb{Z}_3^* \bmod 3^\ell$ ” means hitting all $2 \cdot 3^{\ell-1}$ invertible residue classes.

Conjecture 7.1 (Weak Covering Conjecture; Wirsching 1998, Conj. 3.9). *There is a sub-exponential function $K(\ell) > 0$ (i.e. $\lim_{\ell} K(\ell)e^{-\gamma\ell} = 0$ for every $\gamma > 0$) such that, for all j, ℓ : $|R_{j-1,j}| \geq K(\ell) \cdot 3^\ell \Rightarrow R_{j-1,j}$ covers $\mathbb{Z}_3^* \bmod 3^\ell$.*

For $j \geq \ell$, terms of index $i \geq \ell$ vanish mod 3^ℓ , giving the useful reduction $R_{j-1,j} \bmod 3^\ell = 2^{j-\ell} \cdot R_{\ell-1,j} \bmod 3^\ell$, which cuts the enumeration cost from $\binom{2j}{j}$ to $\binom{j+\ell}{\ell}$.

7.2 Implementation and validation

We implemented a cyclic-rotation bitset dynamic program: processing candidate exponents in decreasing order, we maintain, for each count c of terms chosen so far, the set of reachable partial sums mod 3^ℓ as a single Python integer bitset; choosing the exponent v at position c is a cyclic rotation by $2^v \cdot 3^c \bmod 3^\ell$, and population count of the final bitset gives the image size directly. The implementation matched an independently computed reference table exactly ($j^*(\ell)$ for $\ell = 1, \dots, 7$: 1, 4, 6, 7, 9, 10, 11) and a pointwise check ($\ell = 2, j = 2$: image = $\{1, 2, 5, 7\} \bmod 9$, missing $\{4, 8\}$).

7.3 Results

We computed $j^*(\ell)$, the smallest j for which $R_{j-1,j}$ covers, for $\ell = 1, \dots, 20$ (cost grew by a factor ≈ 3.3 per step near the end; $\ell = 20$ alone took ≈ 27 minutes; further extension in pure Python was judged not worthwhile). Writing $e(\ell) := j^*(\ell) - \log_4 3 \cdot \ell$ for the excess over the entropy-matched threshold $\log_4 3 \approx 0.7925$:

ℓ	j^*	j^*/ℓ	$e(\ell)$
1	1	1.0000	0.208
5	9	1.8000	5.038
10	15	1.5000	7.075
15	20	1.3333	8.113
16	20	1.2500	7.320
17	21	1.2353	7.528
18	22	1.2222	7.735
19	23	1.2105	7.943
20	24	1.2000	8.150

(Full table for all 20 values is in the accompanying computational record.) $j^*(\ell)$ exists and is finite throughout the tested range. An earlier reading of the first 17 points suggested a monotonically decelerating local slope approaching the threshold $\log_4 3$, which the three new points at $\ell = 18, 19, 20$ (all increments exactly $+1$, no new plateaus) revealed to be an artifact of an isolated plateau at $\ell = 16$ crossing the sliding measurement window: without it, the local slope oscillates in 0.74 – 0.86 , straddling $\log_4 3$ from both sides rather than converging cleanly.

Model comparison on the tail $\ell = 10$ – 20 (constant vs. logarithmic vs. square-root vs. slow-linear growth of $e(\ell)$, by AIC) now disfavors pure stabilization ($\Delta\text{AIC} \approx 5$), consistent with a plateau-frequency test (only 1 plateau observed in 19 increments versus ≈ 3.9 expected under stabilization at rate $\log_4 3$; binomial $p \approx 0.072$, marginal but in the same direction). The four slow-growth models remain statistically indistinguishable from one another in this range ($\Delta\text{AIC} < 0.5$); they would only separate by ≥ 1 unit of j^* near $\ell \approx 40$, out of reach of this algorithmic approach.

Theorem 7.2 (Direct computational test of the Weak Covering Conjecture). *$j^*(\ell)$ exists and is finite for all $\ell \leq 20$ (the first direct computational test, to our knowledge, of the exact object of Conjecture 7.1). The excess $e(\ell)$ qualitatively favors unbounded slow growth over stabilization (two independent statistics, ΔAIC and plateau frequency, in the same direction but not individually decisive), consistent with the weak form (Conjecture 7.1) and disfavoring the corresponding strong form with $K(\ell)$ bounded.*

8 Three precision regimes for Tao’s bivariate extension

8.1 Why the naive bivariate argument fails

A natural strategy for closing the gap of Theorem 4.3 is to apply Proposition 1.1 twice, once per sibling leaf, and combine via some form of conditional independence given the common ancestor. A close reading of the mechanism of Tao’s Section 7 proof (characters χ_ξ , i.i.d. Pascal-distributed valuation pairs, a phase evolving multiplicatively by $\log 9 / \log 2$, a “black set” of well-separated triangles, a two-dimensional renewal process with monotonicity) shows that this mechanism operates entirely on $\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ as constructed from *independent* geometric digits by definition (1); the link to actual integers is made elsewhere in [1] (via a total-variation approximation valid only for depth $n \lesssim \log N$).

For our problem — a single fixed integer v , two sibling leaves $w_1(v), w_2(v)$ — the two leaves are *not* two i.i.d. samples; they are two different deterministic functions of the *same* variable v .

Proposition 8.1 (Perfect dependence at fixed depth). *For a fixed pair of sibling paths of depth D , the existence congruence pins v to a single class mod 3^D : writing $v = v_0 + 3^D t$, the two leaves become affine functions of the same free variable t , with unit multipliers: $w_i = w_i(v_0) + 2^{A_i} t$.*

Consequently the pair $(w_1, w_2) \bmod 3^\ell$ lies on a line of size 3^ℓ inside $(\mathbb{Z}/3^\ell\mathbb{Z})^2$, for every $\ell \geq 1$ — perfect deterministic dependence at every precision, with an entire line of resonant frequencies where the correlation has modulus exactly 1.

Consequently, “apply Proposition 1.1 twice, assuming conditional independence given the ancestor” is not merely circular: it is false as stated, at any precision.

8.2 The correct, non-circular reformulation

The quantity that actually matters — decorrelation of *subtree aggregates*, not leaf-to-leaf — is a second-moment statement, expanding into a sum over *pairs of paths*:

$$\mathbb{E}_v[Z_1 Z_2] \propto \#\{(a, a') : a_1 \in B_1, a'_1 \in B_2, \text{Syrac}(a) \equiv \text{Syrac}(a') \bmod 3^D\},$$

which by Fourier inversion gives $\text{Cov} \propto \sum_{\xi \neq 0} S_1(\xi) S_2(\xi)^*$, with S_i character sums conditioned on the first step. Here there is no circularity: the two path indices are independent by the tautology of expanding a square, not by an arithmetic hypothesis.

8.3 The three regimes

1. **Fixed pair, any ℓ :** perfect dependence (Proposition 8.1); no decay is possible, and decorrelation can only come from averaging over paths, never leaf-to-leaf.
2. $\ell = O(\log D)$: *provable*, with Tao’s Section 7 machinery essentially unchanged. The budget $|\rho(\xi)| \leq C_A D^{-A}$ closes the sum over $3^\ell - 1$ frequencies provided $3^\ell \ll D^{2A}$, i.e. polynomial moduli in D — an honest, reachable lemma, though its most natural formulation turns out to be false as originally sketched (§10).
3. $\ell \asymp c \cdot D$ (where the endogeneity barrier, the fresh digits, and Wirsching’s Weak Covering Conjecture all live): this requires *exponential* (power-saving), ξ -uniform Fourier decay; Proposition 1.1 gives only super-polynomial decay, and Tao’s Section 7 method structurally cannot give more, because the losses in the black-set argument are tied to Diophantine properties of $\log 3 / \log 2$ via the slope $\log 3 / \log 2$.

Theorem 8.2 (Regime 3 is a recognized open problem). *Regime 3 above is equivalent, in difficulty, to (i) correlations of digital functions in bases 2 and 3 [20], (ii) effective rigidity for $\times 2, \times 3$ -invariant structures in the sense of Furstenberg (the pair (w_1, w_2) is v observed through two elements of the affine semigroup generated by $x \mapsto (2^a x - 1)/3$), open in the generality required here, and (iii) exponential Fourier decay for 3-adic self-similar measures of Breuillard–Varjú type, known only in special algebraic cases. None of these furnishes a usable tool today.*

9 Triangulation: three independent formulations of the same obstruction

9.1 $\beta = 1$ (Tao 2020) is the Weak Covering Conjecture (Wirsching 1998)

Tao [3] defines $c_n := \inf_{b \in \mathbb{Z}/3^n\mathbb{Z}, 3 \nmid b} \Pr(\text{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) = b)$ and proposes the conjecture $\beta = 1$: $c_n = 3^{-n+o(n)}$ (“no residue class has probability much smaller than average”), proving conditionally that $\beta = 1$ implies Collatz density $\gg x^{1-o(1)}$ (almost total), and noting that the best known unconditional bound is Krasikov–Lagarias’ $\gamma = 0.84$ [14].

Proposition 9.1 (Structural identity). *After a unit twist, Tao’s Syracuse random variable and Wirsching’s sets $R_{j,k}$ are the same algebraic object (sums of powers of 2 and 3 with strictly decreasing exponents). Consequently, Conjecture 7.1 (sub-exponential $K(\ell)$) implies $\beta = 1$ (an exact entropy count: $j = \log_4 3 \cdot \ell + o(\ell)$ gives $c_\ell \gtrsim 3^{-\ell(1+o(1))}$); conversely $\beta = 1$ alone gives only an exponentially weakened form of Conjecture 7.1. The two are sibling formulations on the same exact entropy scale — the threshold $\log_4 3 \approx 0.7925$ of §7 is the same identity 4^j vs. 3^ℓ on both sides, not a coincidence.*

Corollary 9.2. *The excess $e(\ell) = j^*(\ell) - \log_4 3 \cdot \ell$ measured in §7 is literally the $o(n)$ term of $\beta = 1$ ($c_\ell \gtrsim 3^{-\ell} \cdot 4^{-e(\ell)}$): our qualitative reading of $e(\ell)$ (sub-linear growth favored) is direct computational evidence in favor of $\beta = 1$, not merely of Conjecture 7.1.*

Tao’s 2020 approach also shows that the endogeny/decorrelation route of §4 is not the only path: it replaces “near-independence between subtrees” with “deterministic separation of preimages (easy) plus L^∞ marginal control ($\beta = 1$, where all the difficulty lives)”. This confirms that a route avoiding decorrelation altogether exists, and that it hits the *same* wall in marginal form.

9.2 Wirsching’s 2003 functional equation and the base-3 Fabius function

A different, more refined route toward the same goal is Wirsching [5], whose central combinatorial object, the “Elka” functions $e_\ell(k, a) := |E_{\ell,k}(a)|$, count paths of length $k + \ell$ in the Collatz graph ending at a — the same counting object underlying our entire $G(v)$ /Syrac program, now viewed on the state space $\mathbb{X} = \mathbb{I}_3 \times \mathbb{Z}_3^\times$: the $\mathbb{Z}_3^\times$ factor carries the 3-adic variable (habitat of the Syracuse measure), while the $\mathbb{I}_3 \cong \{0, 1, 2\}^\mathbb{N}$ factor carries the normalized step-count $k/3^\ell$, converging to a density φ .

Proposition 9.3 (Base-3 Fabius function). *φ is the density of $X = \sum_{j \geq 1} 2U_j \cdot 3^{-j}$, U_j i.i.d. uniform on $[0, 1]$ — the base-3 analogue of the Fabius function (which satisfies $f'(x) = 2f(2x)$; here $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $[0, 2/3]$), a member of the family of “atomic functions” of Rvachev. It is the unique $L^1([0, 1])$ fixed point of the averaging operator $W_3 f(x) := \frac{3}{2} \int_{3x-2}^{3x} f(t) dt$, C^∞ , piecewise polynomial away from the standard Cantor set, with certified geometric convergence $\|f_n - \varphi\|_1 \leq 2^{-n+1} \|f_1 - f_0\|_1$.*

Wirsching’s chain of implications reduces uniform positive predecessor density to a sequence of three conjectures, the most concrete of which (Conjecture 3, an inequality on $\liminf \varphi(z_\ell)/\varphi_0(z_\ell)$ along sequences in the central-limit window $|\ell - k_\ell| \leq \delta\sqrt{\ell}$) we tested numerically.

9.3 Certified numerical test of Wirsching’s Conjecture 3

Using the exact self-similarity $X \stackrel{d}{=} (2U + X)/3$, the moments $M_i = \mathbb{E}[X^i]$ satisfy the exact rational recursion

$$M_i = \frac{1}{3^i - 1} \sum_{k=1}^i \binom{i}{k} \frac{2^k}{k+1} M_{i-k},$$

and, combined with an antiderivative reduction expressing φ at extreme-tail arguments in terms of exact binomial moment sums (rather than iterating W_3 , whose L^1 error certificate is too weak to control pointwise values at the relevant scale), we obtain a zero-truncation-error evaluation of φ at depth ℓ up to $\ell = 500$ (working throughout in log-space with 60-digit precision; error certified $\leq 10^{-8}$).

Theorem 9.4 (Numerical test of Wirsching’s Conjecture 3). *The decisive ratio $L_\ell := 3^{1-\ell}\varphi(3x_\ell^+)/\varphi(x_\ell^+)$, measured for $\ell = 250, \dots, 500$ in the CLT window ($u \in [-2, 2]$), converges to the predicted value $2/3$ with deficit $\ell \cdot (2/3 - L_\ell) \rightarrow 0.580 \pm 0.001$ — a coefficient independently reproduced by Berg–Krüppel’s own asymptotic φ_0 . The quantity $\ln(\varphi/\varphi_0)$ is increasing, concave, and uniform across $u \in [-2, 2]$ (dispersion $< 10^{-4}$), consistent with a finite limit $L = -0.619 \pm 0.001$ (stat), with a cauda-shape systematic uncertainty of ± 0.015 , giving $c = e^L \approx 0.538$ (honest range 0.53–0.55), best fit by a cauda term $0.79/\sqrt{\ell}$; no log-periodic modulation is detected. This supports Conjecture 3 (Wirsching’s most concrete conjecture in the chain) with certified numerical error, on the tested range; Conjectures 1 and 2 above it remain open.*

Together, Theorems 7.2, Proposition 9.1, and 9.4 give three independent formal “roupagens” — integer counting (Wirsching 1998), Fourier characters (Tao 2020), a real functional equation (Wirsching 2003) — of the same critical window, none resolved.

10 Two candidate lemmas, executed to completion

Sections 8–9 identify two natural next steps: a “regime-2 lemma” (decorrelation of subtree aggregates at logarithmic precision) and a “Z-number reduction lemma” (formalizing the analogy with Littlewood–Offord/Bohr-set methods from [2]). We executed both to completion; neither survives as originally sketched, but both yield genuine, citable results.

10.1 The regime-2 lemma: refutation, structure theorem, and a computational refutation of its natural sufficient condition

The originally sketched budget, $|\rho(\xi)| \leq C_A D^{-A}$ uniformly in ξ , contradicts Proposition 1.2 (an exact identity, not a lossy estimate): for any frequency ξ of primitive level ℓ' (i.e. $\xi = 3^{\ell-\ell'}\xi_0$, $3 \nmid \xi_0$) and any depth $D \geq \ell'$, $\mathbb{E}[e(\xi F_D/3^\ell)] = \varphi_{\ell'}(\xi_0)$ is independent of D : depth beyond ℓ' is a dummy parameter, because 3-adic ultrametricity truncates the relevant sum to the coarsest ℓ coordinates (Proposition 1.2). A closed example at $\ell' = 1$: $\text{Syrac}_1 = 2^{-a} \bmod 3 \in \{1, 2\}$ with probabilities $1/3, 2/3$; $|\varphi(1)| = 1/\sqrt{3} \approx 0.577$ for every D , with no decay whatsoever. The correct per-frequency budget is $C_A \ell'^{-A}$ (Proposition 1.1, which requires $3 \nmid \xi$), not $C_A D^{-A}$; even with this correction, the total ℓ^∞ budget summed over frequencies, $\sum_{\ell' \leq \ell} 3^{\ell'} (C_A \ell'^{-A})^2 \asymp 3^\ell \ell^{-2A}$, never tends to 0: closing by frequency counting would require square-root cancellation, $\ell^{-\ell'/2-\varepsilon}$ decay — Regime 3 resurfacing inside what should have been the easy regime.

What survives is a genuine structure theorem. With $w_2 = 2^\Delta w_1 + c_\Delta$, $c_\Delta = (2^\Delta - 1)/3$ (a unit multiplier mod 3^ℓ) relating admissible sibling leaves at exponent gap Δ :

Lemma 10.1 (Covariance spectral identity). *For subtree-aggregate observables $Z_i := N_{\eta_i}^{(m)}(w_i(v))$ (weighted populations truncated to precision m), with v uniform on admissible classes mod 3^{m+1} ,*

$$\text{Cov}(Z_1, Z_2) = 3^{-2m} \sum_{\xi \neq 0 \bmod 3^m} e(-\xi c_\Delta/3^m) S_1(-2^\Delta \xi) S_2(\xi),$$

by character orthogonality and the affine sibling relation. The two summation indices are independent by the tautology of expanding the square (not an arithmetic hypothesis), but the dependence through the common v is not thereby cancelled: it survives entirely as the diagonal frequency pairing $\xi_1 = -2^\Delta \xi_2$.

Proposition 10.2 (Exact coarse endogeny). *For complete populations ($\eta_i \equiv 1$) and precision $\ell = 1$: $\text{Corr}(Z_1, Z_2) = +1$ if $\Delta \equiv 0 \pmod{6}$, and $= -1/2$ otherwise — independent of depth D and of any truncation.*

Proposition 10.2 converts the endogeny barrier of Theorem 4.3 from a qualitative argument into a closed form: aggregating leaves does *not* decorrelate them; aggregation projects onto the coarse σ -algebra generated by $w_i \bmod 3^\ell$, where the affine sibling relation of the fixed-pair regime (§8, Regime 1) reappears intact rather than vanishing. This is a positive result with the opposite sign from what was hoped for: what is provable is the persistence of coarse correlation, not its disappearance.

Decomposing the sum of Lemma 10.1 by primitive level and applying Cauchy–Schwarz plus Parseval gives $\text{Cov} = C_{\text{exact}}(\ell_0; \Delta) + \text{Err}$, with C_{exact} exact, finite, and D -independent (Proposition 10.2), and

$$\frac{|\text{Err}|}{\mathbb{E}[Z_1]\mathbb{E}[Z_2]} \leq K_m - K_{\ell_0}, \quad K_\ell := 3^\ell \sum_y \mu_\ell(y)^2,$$

the normalized *collision mass* of the Syracuse measure μ . A conditional lemma would follow if $K_\infty := \lim_\ell K_\ell < \infty$ (equivalently, the density $f = d\mu/d(\text{Haar})$ lies in $L^2(\mathbb{Z}_3)$): the fine covariance, net of the exact coarse component, would then be $\leq K_\infty - K_{\ell_0} \rightarrow 0$, uniformly in m, D, Δ .

Theorem 10.3 (Computational refutation of the L^2 hypothesis). *K_ℓ was computed exactly via the recursion*

$$\mu_n(y) = \frac{1}{2} \nu(2y \bmod 3^n) + \frac{1}{2} \mu_n(2y \bmod 3^n), \quad (7)$$

where ν is the law of $1 + 3F_{n-1}$ (derived below), solved as a truncated geometric series along the cyclic orbit of multiplication by 2 mod 3^n (a single orbit, since 2 is a primitive root mod 3^n for every n). For $\ell = 1, \dots, 17$, K_ℓ grows cleanly and consistently linearly, with increments converging monotonically to ≈ 0.47 , with no sign of saturation ($K_1 = 5/3$ exactly, matching a hand computation from $\Pr(\text{Syrac}_1 = 1) = 1/3$, $\Pr(\text{Syrac}_1 = 2) = 2/3$). Hence $K_\infty = \infty$: the L^2 hypothesis fails.

Derivation of the recursion (7). By Proposition 1.2, $F_n(a) = 2^{-a_1}(1 + 3F_{n-1}(a_2, \dots, a_n)) \bmod 3^n$. Write $X := 1 + 3F_{n-1}(a_2, \dots, a_n)$, with law ν , independent of a_1 . Conditioning on $a_1 = 1$ (probability 1/2) gives $F_n = 2^{-1}X$; conditioning on $a_1 \geq 2$ (probability 1/2), the memorylessness of the geometric distribution gives $a_1 - 1 \sim \text{Geom}(2)$ again, so $F_n = 2^{-1}(2^{-(a_1-1)}X)$, and the bracketed quantity has exactly the law of F_n itself (an independent copy). This gives the self-referential fixed point $\mu_n(y) = \frac{1}{2}\nu(2y) + \frac{1}{2}\mu_n(2y)$, all arithmetic mod 3^n . $\square$

We validated the recursion — not merely its $\ell = 1$ base case, which would not exercise the recursive machinery at all — against direct Monte Carlo sampling of Syrac from its primary definition (1) at $\ell = 3, 4$ (3×10^6 samples each), bin-by-bin: maximum discrepancy 0.000311 ($\ell = 3$) and 0.000213 ($\ell = 4$), both within the expected Monte Carlo noise ($\approx 1/\sqrt{N} \approx 0.000577$).

Remark 10.4 (Connection to the tail exponent). H -104 (an earlier stage of this project) measured a tail exponent $\alpha \approx 2$ for subtree-population ratios — exactly the critical exponent of the L^2 condition ($\alpha > 2 \Rightarrow K_\infty < \infty$; $\alpha < 2 \Rightarrow K_\ell$ diverges polynomially; $\alpha = 2$ is the borderline case). Theorem 10.3’s clean linear growth is consistent with α slightly below 2, or exactly at the threshold with a logarithmic correction.

Corollary 10.5 (A ladder of “flatness” for the Syracuse measure). L^1 (proved, Tao’s Prop. 1.14: $\text{Osc}_{m,n} \ll_A m^{-A}$, giving total-variation approximability by absolutely continuous measures) $\succ L^2$

(refuted here, Theorem 10.3) $\succ L^\infty/\beta = 1$ (§9, open) — neither comparable to nor implied by L^2 , but living on the same “beyond total-variation” floor of the ladder.

The regime-2 lemma is thus not an easier stepping stone toward Regime 3: it is a sibling of it, sharing the same missing ingredient, now with a direct computational refutation of the natural sufficient condition rather than mere silence.

10.2 The Z-number reduction: a conditional dichotomy and an exact negative result

The second candidate lemma formalizes “failure of the Weak Covering Conjecture at scale ℓ with margin c implies a Bohr-type concentration configuration”, via the Littlewood–Offord/Riesz-product technique of Tao’s 2011 blog post [2] (not a paper; we flag this bibliographic correction explicitly, since it is a recurring source of miscitation). That post proves conditional separation results for $2^a - 3^k$ via Halász’s method [22] and inverse Littlewood–Offord theory [23, 24], but acknowledges that Baker’s theorem already gives an unconditionally stronger separation, so treats its own propositions as methodological exercises. Mahler’s classical “Z-number” problem [25] concerns $\xi(3/2)^n$, a genuinely different (though structurally adjacent) object from ours (which concerns $\times 2 \bmod 3^s$ at finite depth, not $\times 3/2 \bmod 1$ on an infinite orbit); the honest anchor for our object is the Erdős/Lagarias family [27, 28].

Definition 10.6 (Failure with margin c). For $c > 0$, the Weak Covering Conjecture *fails at scale ℓ with margin c* if, for $j = \lceil (1+c)\log_4 3 \cdot \ell \rceil$, some $b \in (\mathbb{Z}/3^\ell\mathbb{Z})^\times$ lies outside the image of $R_{j-1,j}$. Set $\gamma_c := (j+\ell)/\ell = 1 + (1+c)\log_4 3$.

Definition 10.7 (Diagonal configuration). A configuration of depth s at slope γ , tolerance δ , and exception rate η is $\xi \in \mathbb{Z}$, $3 \nmid \xi$, such that in $\geq (1-\eta)s$ of the scales $t \in \{1, \dots, s\}$ there is $a \in [\gamma t - t^{2/3}, \gamma t + t^{2/3}]$ with $\|\xi \cdot 2^a/3^t\| < \delta$.

Lemma 10.8 (Pre-wrap triviality). For $\xi = 1$, $\|2^a/3^s\| < \delta$ holds trivially for $a \leq \log_2 3 \cdot s + \log_2 \delta$ (the representative has not yet wrapped around). Since $\gamma_c \geq 1.7925 > \log_2 3 \approx 1.585$ for every $c \geq 0$, the forced configuration lives entirely in the non-trivial post-wrap regime, by structural slack coming from the 4^j vs. 3^ℓ entropy accounting.

Definition 10.9 (Spectral concentration $SC(\varepsilon)$). $\mu_{\ell,j}$ (the law of $Y = \sum R \bmod 3^\ell$) satisfies $SC(\varepsilon)$ if there is $\xi \not\equiv 0 \bmod 3^\ell$ with $|\hat{\mu}_{\ell,j}(\xi)| \geq 3^{-\varepsilon\ell}$.

Lemma 10.10 (Conditional reduction). For $c > 0$ and $\varepsilon < \varepsilon_0(c)$ there exist $\delta(\varepsilon, c) < 1/2$, $\eta(\varepsilon, c) < 1$ such that: failure at scale (ℓ, c) together with $SC(\varepsilon)$ implies existence of a (γ_c, δ, η) -diagonal configuration of depth $s \geq (1 - O(\varepsilon))\ell$. Contrapositively: absence of diagonal configurations of slope γ_c at large depth implies any failure of the Weak Covering Conjecture, if it exists, is spectrally diffuse ($SC(\varepsilon)$ fails for every ε).

The proof (a six-step argument: Fourier-inversion lower bound on $\sum |\hat{\mu}(\xi)|$, transfer to a tilted i.i.d. model at parameter $p_c = 1/\gamma_c$, extraction of alternable parallelepipeds via the Riesz-product device of [2], Halász’s theorem, a routine scale descent, and an upgrade from a diagonal configuration to a continuous segment) is complete except for the last step, which requires stability of the relevant frequency ξ under $j' \mapsto j' + 1$ across scales — plausible but not established, and declared as an open gap rather than glossed over.

10.3 Proposition C: an exact wall of constants

Proposition 10.11 (Halász deficit). *Unconditionally, Halász’s theorem gives only $\tau \geq 3^{-\ell}$, not $SC(\varepsilon)$, yielding a budget $\sum \|\cdot\|^2 \leq 0.549\ell$ against a required ceiling of 0.25ℓ ($\delta = 1/2$, $d = \ell$ generators): a deficit of a factor ≥ 2.2 , rising to ≈ 7 at a realistic generator density $\rho \approx 0.3$. In Tao’s original 2011 application this closes because $\log q \ll d$ (unbounded ratio); in the Weak Covering Conjecture regime, $\log_2 q/d \geq 1.6$ — the technique does not merely degrade, it inverts sign.*

Theorem 10.12 (Jensen identity and the exact Fourier-budget deficit). *Let $Z = \sum_{g \geq 1} w_g e(U_g)$, U_g i.i.d. uniform, $w_g = p(1-p)^{g-1}$ ($\sum_g w_g = 1$). Then, exactly,*

$$\Lambda := \mathbb{E}[\log(1/|Z|)] = \log(1/p). \quad (8)$$

Proof. By the memoryless self-similarity of the geometric weights ($w_g/(1-w_1) = p(1-p)^{g-2}$ for $g \geq 2$, the same geometric sequence re-indexed), $Z \stackrel{d}{=} p e(U_1) + (1-p) Z'$, with Z' an independent copy of Z ; since $|Z'| \leq 1$ always (a convex combination of unimodular terms), and $p/(1-p) \geq 1$ for $p \geq 1/2$, we have $|Z'| \leq p/(1-p)$ whenever $p \geq 1/2$. Writing $c := (1-p)Z'/p$, so $|c| \leq 1$, factor $p e(U_1) + (1-p)Z' = p e(U_1)(1 + c e(-U_1))$ and take $\log|\cdot|$: the function $z \mapsto \log|1 + cz|$ is (the real part of) a holomorphic function of z with no zero inside the disc $|z| < 1/|c| \ni$ the unit circle (as $|c| \leq 1$), so by the mean-value property (Jensen’s formula with no zeros enclosed) its average over $z = e(U_1)$, U_1 uniform, equals its value at $z = 0$, namely 0. Hence $\mathbb{E}_{U_1} \log|p e(U_1) + (1-p)Z'| = \log p$ for every realization of Z' , and averaging over Z' gives $\mathbb{E}[\log|Z|] = \log p$, i.e. (8). $\square$

Applying Theorem 10.12 with $p = p_c = 1/\gamma_c$ (the tilted model matching the Weak Covering Conjecture’s threshold slope) gives $\Lambda = \log \gamma_c \approx 0.5834 + O(c)$, confirmed by Monte Carlo (8×10^6 samples: 0.58344 ± 0.0005), against the value $\log 3 \approx 1.0986$ that would be needed to exclude a diffuse hole by an annealed ℓ^1 Fourier-budget argument: a deficit factor of ≈ 1.88 . The annealed criterion would only invert at $\gamma = 3$ ($j \geq 2\ell$), far above the regime of the conjecture ($j^*(\ell)/\ell \approx 1.2$, Theorem 7.2).

Theorem 10.13 (Proposition C: self-consistency of a diffuse hole). *A spectrally diffuse failure of the Weak Covering Conjecture satisfies the annealed ℓ^1 Fourier budget with slack ≈ 1.88 ; no Littlewood–Offord/Riesz-product counting argument of the kind used in [2] can exclude it. This branch — inaccessible to ℓ^1 methods by Theorem 10.13 — is exactly the branch identified as the core of $\beta = 1$ (§9) and of the refuted L^2 condition (Theorem 10.3): a fourth and fifth independent articulation of the same missing ingredient.*

We emphasize the correct framing, consistent with the fragilities listed above: this is a reduction of the spectrally *concentrated* branch to the Erdős–Lagarias–Furstenberg family, together with a proof that the spectrally *diffuse* branch is inaccessible to ℓ^1 Littlewood–Offord methods — not a reduction of the Weak Covering Conjecture to the classical Z-number problem, which Theorem 10.13 and the discussion above show would be imprecise.

11 Contextualizing the barrier in the wider literature

A directed literature search, going beyond the Collatz-specific literature into the general theory of Fourier decay for self-similar measures, situates the barrier precisely and rules out several plausible-looking shortcuts, each for a specific, verified reason.

11.1 The Syracuse measure is a rigid 3-adic self-similar measure

Unrolling the renewal recursion of (1) gives, in the limit,

$$X = \sum_{k \geq 1} 3^{k-1} 2^{-S_k}, \quad S_k := a_1 + \cdots + a_k, \quad (9)$$

the fixed point of the countable affine iterated function system $\phi_a(x) = 2^{-a}(1+3x)$ on $\mathbb{Z}_3$, weighted by $\Pr(a) = 2^{-a}$, every map sharing the *same* ultrametric contraction ratio $|3 \cdot 2^{-a}|_3 = 1/3$. This is a genuine (if non-standard, countable, overlapping) self-similar structure, so the question of whether the general theory of Fourier decay for self-similar measures [15, 16, 17] transfers is a fair one to ask, and we checked it explicitly.

Proposition 11.1 (Rigidity of $\mathbb{Z}_3^\times$). *Every infinite closed subgroup of $(\mathbb{Z}_3, +)$ has finite index (it equals $3^k \mathbb{Z}_3$ for some $k \geq 0$); consequently, since $\mathbb{Z}_3^\times \cong \mathbb{Z}/2 \times \mathbb{Z}_3$, every infinite closed subgroup of $\mathbb{Z}_3^\times$ has finite index. Since 2 is a topological generator of $\mathbb{Z}_3^\times$ (a primitive root modulo 3^n for every n), the unit phases 2^{-a} spread maximally, the opposite of the “thin multiplicative orbit” phenomenon (a Pisot-type obstruction to the Rajchman property) that drives non-uniform Fourier decay on the real line.*

Theorem 11.2 (The general theory does not transfer). (a) *The pushforward mechanism of Baker–Banaji [15] (Archimedean stationary phase, requiring a C^2 map with $f'' \neq 0$ off a null set) would give, at best, qualitative Rajchman decay for the analogue considered here — strictly weaker than the super-polynomial decay already guaranteed by Proposition 1.1.*

(b) *Brémont’s Pisot-type criterion for non-Rajchman self-similar measures, transported to $\mathbb{Z}_3$, trivializes by Proposition 11.1: there is no room for a “thin” 3-adic multiplicative orbit of the kind the criterion requires.*

(c) *The renewal-theoretic technique of Li–Sahlsten [17], which needs two distinct contraction ratios with a Diophantine relation between their logarithms, also degenerates: every branch of (9) has the same 3-adic contraction ratio $1/3$, so the relevant “scale walk” is deterministic ($S_n = n \log 3$, a pure lattice walk with no overshoot fluctuation), not the non-lattice random walk the technique is built to exploit.*

We stress that Baker–Banaji’s general phenomenon — Rajchman measures with no uniform decay rate are generic, not pathological, in the class of self-similar/self-conformal measures — remains a fair piece of context: it shows that the kind of failure exhibited by Theorem 10.13 has real structural precedent in the reference literature, legitimizing the framing that a proof of $\beta = 1/\text{the Weak Covering Conjecture}$ genuinely needs the specific arithmetic input of this problem, not generic smoothness. It should not, however, be over-interpreted as evidence *for* the diffuse scenario itself: Baker–Banaji’s construction requires fine-tuning of a continuous parameter (a Liouville-type construction, dense in the family but exceptional in measure; typical parameters give polynomial decay [18, 19]), while the Syracuse measure is a rigid arithmetic object with no tunable parameter, and Proposition 11.1 shows the enabling mechanism (fine Diophantine approximation) simply does not exist on the 3-adic side — which cuts against the diffuse scenario as much as for it. We cite Theorem 11.2 with this disanalogy stated explicitly.

11.2 A seventh independent formulation, empirically tested

Very recent independent work [21] reduces the Collatz conjecture, by a purely combinatorial route with no Fourier analysis whatsoever, to a residue-balance condition along a sparse subsequence of

the true (not ensemble) orbit: writing $X_t := \mathbb{K}[n_t \equiv 1 \bmod 4]$ for the compressed odd-to-odd orbit and identifying “burst-ending times” (the last index of a maximal run of $X_t = 1$), that work proves a Map Balance Theorem (the map-level bias between gap-length classes is exactly ± 1 , for every $K \geq 5$) and reduces the conjecture, in the dominant residue class, to a claim that the fraction of burst-endings landing in one of two residues mod 32 stays bounded near $1/2$ along the orbit.

We tested this condition directly on real orbits. An ensemble of many moderate orbits ($2\text{--}5 \times 10^4$ roots, 20–50 bits) shows the aggregate deviation plateauing at 1.2–1.9%, which is on its own consistent with the conjecture (only boundedness is claimed, not convergence to zero); a more faithful test — a single very long orbit (a random 16000-bit start, 38395 compressed steps, 4761 relevant burst-endings) — shows the deviation tracking the $1/\sqrt{i}$ statistical-noise curve closely, with no systematic trend, ending at a deviation of 0.00053. Five further orbits from 500 to 8000 bits show the same pattern. We interpret the ensemble plateau as an artifact of aggregating many short, finite, similarly-phased orbits, and the single-long-orbit result as qualified empirical support for the conjecture of [21], on the scale tested.

This gives a seventh, technically unrelated, articulation of the same missing ingredient: equidistribution/balance of residues along the true orbit, appearing independently in endogeny (§4), the Weak Covering Conjecture (§7), $\beta = 1$ (§9), the L^2 condition (§10), the spectral dichotomy (§10), and now a purely combinatorial reformulation.

12 Discussion: why a precisely characterized barrier is a genuine contribution

A natural objection to the material of §§4–11 is that neither candidate lemma of §10 produced a proof, so the paper might seem to document an absence rather than a result. We think this objection, while natural, does not survive scrutiny of what was actually produced.

First, the outcome of executing both candidate lemmas was not silence: it was two independent, rigorous negative/structural results, each of which pins down *precisely* why a natural approach fails — Proposition 10.2’s exact closed-form persistence of coarse correlation; Theorem 10.3’s direct computational refutation of a specific sufficient condition; Theorem 10.12’s exact identity quantifying, to a specific constant factor (≈ 1.88), why a whole family of Fourier-analytic techniques cannot close the gap. Precisely characterizing an obstruction — showing not merely that an approach fails, but *exactly* why, with a quantified margin — is itself a recognized and citable form of mathematical contribution, independent of whether the underlying conjecture is eventually resolved.

Second, these results do not stand alone: they converge, by seven technically unrelated routes (endogeny, the Weak Covering Conjecture, $\beta = 1$, Wirsching’s 2003 functional equation, the L^2 condition, the spectral dichotomy of Proposition C, and the purely combinatorial reformulation of [21]), on what appears to be a single underlying arithmetic statement: some form of equidistribution or near-independence of 3-adic digits along the true, deterministic integer dynamics, not merely along an idealized i.i.d. model. A barrier that is only visible from one angle is a natural candidate for being an artifact of that angle’s specific technique; a barrier that recurs, in essentially the same form, across seven independent technical vocabularies (character sums, smoothing transforms, a real functional equation, Jensen’s identity, a self-similar-measure classification, and elementary residue counting) is much better evidence that the obstruction is a structural feature of the problem itself.

Third, this theoretical package sits alongside a result (Theorem 6.1) whose validity does not depend on any of the above framing at all: a direct, falsifiable, statistically calibrated empirical measurement resolving a fifteen-year-old dispute in the literature. We consider this the single most framing-independent contribution of the paper, and recommend it be weighted accordingly by a

reader assessing the paper’s overall reliability.

We therefore present this paper’s contribution not as an attempted proof of any part of the Collatz conjecture, nor as a report of failure, but as a precise map of where a specific, natural, technically rich line of attack — Tao’s Syracuse-measure program, generalized to $qx + 1$ — runs into a wall, together with an exact quantification of the size and shape of that wall from six independent angles, and one hard, unconditional empirical result obtained along the way.

13 Conclusion and open directions

We summarize the concrete open problems this paper leaves in the cleanest form we have been able to give them.

- (O1) **Near-independence of fresh 3-adic digits across sibling subtrees** (Theorem 4.3): the arithmetic statement whose idealized (i.i.d.) analogue is provable via smoothing-transform classification and Choquet–Deny rigidity.
- (O2) **Wirsching’s Weak Covering Conjecture** (Conjecture 7.1), equivalently (Proposition 9.1) Tao’s $\beta = 1$ (§9), open since 1998 and 2020 respectively.
- (O3) **Wirsching’s Conjectures 1 and 2** (§9), above the numerically-tested Conjecture 3, of the same central-limit-window difficulty.
- (O4) **Exponential, ξ -uniform Fourier decay for the Syracuse measure at depth $\ell \asymp D$** (§8, Regime 3), equivalent to recognized open problems in effective $\times 2, \times 3$ rigidity and Fourier decay of self-similar 3-adic measures.
- (O5) **Exclusion of spectrally diffuse Weak Covering Conjecture failures** (Theorem 10.13), shown here to be inaccessible to ℓ^1 Littlewood–Offord methods, leaving genuinely new technique as the only route.
- (O6) **Chang’s residue-balance conjecture** [21] (§11), supported empirically here on a single very long orbit, unproved.

Any progress on any one of (O1)–(O6) would very likely transfer, given the structural identities established in this paper, to progress on several of the others simultaneously. We regard this convergence as the paper’s central finding.

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of Theorem 2.5 attributed to [16] all matched exactly), and [8] was corrected after this process revealed that its title, as originally reconstructed from an AI consultation, was simply wrong. One citation, [22], remains genuinely unresolved: the year originally attached to it (1977) was itself an error (the correct year is 1971), and an extensive search — including a dedicated attempt to locate an open digitization (HathiTrust, the Internet Archive, JSTOR, and Hungarian institutional archives) — did not turn up the primary 1971 text; only a 1972 erratum/remarks note by the same author was obtained. The bibliographic entry itself (title, year, journal, volume, and pages) is independently corroborated by multiple secondary listings (library catalogs and citing articles), so we retain the citation with the corrected year, while flagging that the primary text has not been read by us and recommending a reader verify the specific technical content independently before relying on it. We consider the whole episode — one wrong year caught, one wrong title caught, one theorem statement confirmed verbatim, and one primary source that simply could not be located — good evidence for why this verification pass was worth doing, and an honest illustration of its limits. We follow, in this disclosure, what we believe is best practice in the small existing literature on AI-assisted contributions to this specific problem.

A Numerical validation methodology

We record, for completeness, the validation discipline applied to each computational result reported above, since an unvalidated recursive computation is a known source of silent error.

- *Pressure equation roots* (§3): solved by bisection independently of the closed-form derivation; cross-checked against empirical counting slopes on real reverse trees for $q = 5, 7$.
- *Weak Covering Conjecture DP* (§7): validated against an independently computed reference table for $\ell = 1, \dots, 7$ and a hand-checked point ($\ell = 2, j = 2$).
- *K_ℓ recursion* (Theorem 10.3): the only initially available check was the hard-coded base case ($K_1 = 5/3$), which does not exercise the recursive machinery; we additionally validated the full distribution μ_ℓ , bin by bin, against direct Monte Carlo sampling of the primary definition (1) at $\ell = 3, 4$, within Monte Carlo noise.
- *Jensen-identity Monte Carlo* (Theorem 10.12): the identity is exact and independent of simulation; the Monte Carlo estimate (8×10^6 samples) serves only as a sanity check on the analytic value, and matches it to four significant figures.
- *Wirsching 2003 moment recursion* (Theorem 9.4): validated by an independent fixed-point check on an independent grid, agreeing to the grid’s resolution (≈ 6 digits), and by confirming the functional equation (7.12) of [5] numerically up to $\ell = 200$ using an analytic (not numerically differenced) logarithmic derivative.
- *Chang residue-balance test* (§11): validated by confirming that no burst-ending event ever produced a residue outside the two predicted values ($9, 25 \bmod 32$) in either the ensemble or single-orbit experiments, matching the source paper’s structural proposition exactly.

References

- [1] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum of Mathematics, Pi **10** (2022), e12. arXiv:1909.03562.

- [2] T. Tao, *The Collatz conjecture, Littlewood–Offord theory, and powers of 2 and 3*, blog post, <https://terrytao.wordpress.com/2011/08/25/>, 2011.
- [3] T. Tao, *Equidistribution of Syracuse random variables and density of Collatz preimages*, blog post, <https://terrytao.wordpress.com/2020/01/25/>, 2020.
- [4] G. J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, Lecture Notes in Mathematics **1681**, Springer, 1998.
- [5] G. J. Wirsching, *On positive predecessor density in $3n + 1$ dynamics*, Discrete and Continuous Dynamical Systems **9**(3) (2003), 771–787.
- [6] A. V. Kontorovich and J. C. Lagarias, *Stochastic models for the $3x + 1$ and $5x + 1$ problems*, in: *The Ultimate Challenge: the $3x + 1$ Problem* (J. C. Lagarias, ed.), American Mathematical Society, 2010.
- [7] D. Aldous and A. Bandyopadhyay, *A survey of max-type recursive distributional equations*, Annals of Applied Probability **15**(2) (2005), 1047–1110.
- [8] F. Gonçalves, R. Greenfeld, and J. Madrid, *Generalized Collatz maps with almost bounded orbits*, arXiv:2111.06170, 2022. [Verified directly against the archived PDF (title/authors corrected from an earlier, inaccurate reconstruction).]
- [9] R. Durrett and T. M. Liggett, *Fixed points of the smoothing transformation*, Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete **64**(3) (1983), 275–301.
- [10] G. Alsmeyer, J. D. Biggins, and M. Meiners, *The functional equation of the smoothing transform*, Annals of Probability **40**(5) (2012), 2069–2105.
- [11] K. Kolesko and S. Mentemeier, *Fixed points of the multivariate smoothing transform: the critical case*, arXiv:1409.7220.
- [12] C. M. Goldie, *Implicit renewal theory and tails of solutions of random equations*, Annals of Applied Probability **1**(1) (1991), 126–166.
- [13] D. Buraczewski, E. Damek, and T. Mikosch, *Stochastic Models with Power-Law Tails: The Equation $X = AX + B$* , Springer, 2016.
- [14] I. Krasikov and J. C. Lagarias, *Bounds for the $3x + 1$ problem using difference inequalities*, Acta Arithmetica **109** (2003), 237–258.
- [15] S. Baker and A. Banaji, *Self-similar and self-conformal measures with slow Fourier decay*, arXiv:2602.05593, 2026.
- [16] J. Brémont, *Self-similar measures and the Rajchman property*, Annales Henri Lebesgue **4** (2021), 973–1004. arXiv:1910.03463. [Verified: Theorem 2.5, cited in §11, confirmed against the full text.]
- [17] J. Li and T. Sahlsten, *Trigonometric series and self-similar sets*, Journal of the European Mathematical Society **24**(1) (2022), 341–368.
- [18] B. Solomyak, *On the random series $\sum \pm \lambda^n$ (an Erdős problem)*, Annals of Mathematics **142**(3) (1995), 611–625.

- [19] P. Varjú and H. Yu, *Fourier decay of self-similar measures and self-similar sets of uniqueness*, Analysis & PDE **15**(3) (2022), 843–858. arXiv:2004.09358.
- [20] L. Spiegelhofer, *Collisions of digit sums in bases 2 and 3*, Israel Journal of Mathematics **258** (2023), 475–502. arXiv:2105.11173.
- [21] E. Y. Chang, *A structural reduction of the Collatz conjecture to one-bit orbit mixing*, arXiv:2603.25753, 2026.
- [22] G. Halász, *On the distribution of additive and the mean values of multiplicative arithmetic functions*, Studia Scientiarum Mathematicarum Hungarica **6** (1971), 211–233. [Bibliographic data independently corroborated by secondary sources; primary text not located despite a dedicated search of open digitization repositories. Only a related 1972 erratum/remarks note by the same author has been obtained and checked. See Acknowledgments.]
- [23] T. Tao and V. Vu, *Inverse Littlewood–Offord theorems and the condition number of random discrete matrices*, Annals of Mathematics **169** (2009), 595–632.
- [24] H. Nguyen and V. Vu, *Optimal inverse Littlewood–Offord theorems*, Advances in Mathematics **226** (2011), 5298–5319.
- [25] K. Mahler, *An unsolved problem on the powers of $3/2$* , Journal of the Australian Mathematical Society **8**(2) (1968), 313–321.
- [26] L. Flatto, J. C. Lagarias, and A. D. Pollington, *On the range of fractional parts $\{\xi(p/q)^n\}$* , Acta Arithmetica **70**(2) (1995), 125–147.
- [27] P. Erdős, *Some unconventional problems in number theory*, Mathematics Magazine **52**(2) (1979), 67–70. [Verified: located directly in the reference list of [28], item [4].]
- [28] J. C. Lagarias, *Ternary expansions of powers of 2*, Journal of the London Mathematical Society **79**(3) (2009), 562–588.